QIC 890: Quantum Complexity Theory
Problem Set 1 — Solutions

Solution to Problem 1

(a) By the assumption that $A \in \mathrm{NP}$, there exist a polynomial resource bound p and a language $B \in \mathrm{P}$ such that these implications are true.

$$x \in A_{\mathrm{yes}} \implies \big(\exists\, y \in \{0,1\}^{p}\big)\big[\langle x, y\rangle \in B\big]$$
$$x \in A_{\mathrm{no}} \implies \big(\forall\, y \in \{0,1\}^{p}\big)\big[\langle x, y\rangle \notin B\big]$$

Let $q(n) = p(n) + 1$, and define C to be the language consisting of all strings $\langle x, ay\rangle$ such that either $a = 0$ or $\langle x, y\rangle \in B$, where $x \in \Sigma^*$, $y \in \{0,1\}^{p}$, and $a \in \{0,1\}$.

It is clear that q is a polynomial resource bound given that this is true of p. We also have that $C \in \mathrm{P}$ by the assumption that $B \in \mathrm{P}$; a polynomial-time DTM for C can read the first symbol of the second argument given an input pair $\langle x, ay\rangle$ to determine a, accept if it is 0, and otherwise run a polynomial-time DTM for B on the string $\langle x, y\rangle$.

Observe that

$$\big|\{z \in \{0,1\}^{q} \,:\, \langle x, z\rangle \in C\}\big| = 2^{p} + \big|\{y \in \{0,1\}^{p} \,:\, \langle x, y\rangle \in B\}\big|$$

for every $x \in \Sigma^*$. This is because we have $\langle x, 0y\rangle \in C$ for every $y \in \{0,1\}^{p}$, contributing 2^{p} to the sum, along with $\langle x, 1y\rangle \in C$ for every $y \in \{0,1\}^{p}$ satisfying $\langle x, y\rangle \in B$.

Thus, if $x \in A_{\mathrm{yes}}$, then there exists at least one string $y \in \{0,1\}^{p}$ such that $\langle x, y\rangle \in B$, and therefore

$$\big|\{z \in \{0,1\}^{q} \,:\, \langle x, z\rangle \in C\}\big| > 2^{p}.$$

On the other hand, if $x \in A_{\mathrm{no}}$, then there are no strings $y \in \{0,1\}^{p}$ such that $\langle x, y\rangle \in B$, and therefore

$$\big|\{z \in \{0,1\}^{q} \,:\, \langle x, z\rangle \in C\}\big| = 2^{p}.$$

The required implications follow from $\frac{1}{2} \cdot 2^{q} = 2^{p}$ (with the second implication holding with equality).

(b) We will construct a polynomial resource bound p and a language $B \in \mathrm{P}$ satisfying Definition 2.18, given q and C as in the problem statement.

First, we will take $p = q + r$, where $r = q + 2$. (Thus, $p = 2q + 2$, but it is helpful to write $p = q + r$ to make the argument more clear.) Second, we will take B to be the language containing every pair $\langle x, zw \rangle$, where $x \in \Sigma^*$, $z \in \{0,1\}^q$, and $w \in \{0,1\}^r$, such that both $\langle x, z \rangle \in C$ and $w \neq 0^r$. That is,

$$\langle x, zw \rangle \in B \iff \big[(\langle x, z \rangle \in C) \wedge (w \neq 0^r)\big],$$

ranging over all $x \in \Sigma^*$, $z \in \{0,1\}^q$, and $w \in \{0,1\}^r$.

It is evident that p is a polynomial resource bound, given that this is true of q, and we have $B \in \mathrm{P}$ through the same general sort of reasoning as was used to argue $C \in \mathrm{P}$ in part (a).

Consider the number of binary strings $y = zw$ that satisfy $\langle x, y \rangle \in B$, where $z \in \{0,1\}^q$ and $w \in \{0,1\}^r$, for a given string x. This number is given by

$$\big|\{z \in \{0,1\}^q \,:\, \langle x, z \rangle \in C\}\big| \cdot (2^r - 1)$$

as we have $2^r - 1$ choices of $w \neq 0^r$ for which $\langle x, zw \rangle \in B$ for each $z \in \{0,1\}^q$ satisfying $\langle x, z \rangle \in C$.

Assume first that $x \in A_{\mathrm{yes}}$. In this case, we must have

$$\big|\{z \in \{0,1\}^q \,:\, \langle x, z \rangle \in C\}\big| \geq \tfrac{1}{2} \cdot 2^q + \tfrac{1}{2}.$$

This is because the left-hand side is a natural number and $\frac{1}{2} \cdot 2^q$ is either $1/2$ (in case $q = 0$) or a natural number (if q is positive) — so a strict inequality forces the left-hand side to be at least the original right-hand side plus one-half. Therefore,

$$\big|\{y \in \{0,1\}^p \,:\, \langle x, y \rangle \in B\}\big| \geq \frac{1}{2} \cdot (2^q + 1) \cdot (2^r - 1).$$

Now we can substitute $r = q + 2$ to obtain

$$(2^q + 1) \cdot (2^r - 1) = 2^p + 3 \cdot 2^q - 1 > 2^p,$$

and therefore

$$\big|\{y \in \{0,1\}^p \,:\, \langle x, y \rangle \in B\}\big| > \frac{1}{2} \cdot 2^p. \tag{1}$$

Now assume that $x \in A_{\mathrm{no}}$. This case is more direct: the inequality

$$\big|\{y \in \{0,1\}^p \,:\, \langle x, y \rangle \in B\}\big| \leq \frac{1}{2} \cdot 2^q \cdot (2^r - 1) < \frac{1}{2} \cdot 2^q \cdot 2^r = \tfrac{1}{2} \cdot 2^p \tag{2}$$

follows from the implication in the problem statement along with $2^r - 1 < 2^r$.

The inequalities (1) and (2) are both strict, so $A \in \mathrm{PP}$ by Definition 2.18.

Solution to Problem 2

(a) Suppose first that M_A is a DTM that decides the language A in time $O(4^n)$. Define a DTM M_B that does the following on a given input string $w \in \{0,1\}^*$:

1. Check if w takes the form $w = xx$ for some string $x \in \{0,1\}^*$, rejecting if not.
2. Run M_A on x.

The first step can easily be performed in $O(2^n)$ time for $n = |w|$; in fact it can be done in time $O(n)$. If the second step is performed, then it requires $O(4^{n/2}) = O(2^n)$ steps, as it is a computation of M_A (which runs in time $O(4^m)$ on inputs of length m), run on a string of length $n/2$. The running time of M_B is therefore $O(2^n)$. As M_B decides B, it follows that $B \in \mathrm{DTIME}(2^n)$.

(b) Define a function $f : \{0,1\}^* \to \{0,1\}^*$ as

$$f(x) = xx$$

for all $x \in \{0,1\}^*$, which is easily computed in polynomial time. It is immediate from the definition of B that

$$x \in A \Leftrightarrow f(x) \in B$$

and therefore $A \leq_{\mathrm{m}}^{\mathrm{p}} B$.

Solution to Problem 3

Two things must be proved: 1) QCIRCUIT-ACC $\in$ BQP, and 2) every promise problem in BQP Karp reduces to QCIRCUIT-ACC.

For the first, we assume that an input string $\langle Q \rangle$ to QCIRCUIT-ACC is a binary string (though the solution is easily adapted to encodings over any alphabet). Define a family $\{R_x : x \in \{0,1\}^*\}$ of quantum circuits as follows:

$$R_x = \begin{cases} Q & x = \langle Q \rangle \text{ for a quantum circuit } Q \text{ over the standard gate set} \\ & \text{having no input qubits and one output qubit} \\ Q_0 & \text{otherwise,} \end{cases}$$

where Q_0 is a trivial quantum circuit consisting of a single ancillary gate. Note that any input string x satisfying the promise for QCIRCUIT-ACC falls into the

first case; the second case is only needed to guarantee that we have a well-defined family.

The family $\{R_x : x \in \{0,1\}^*\}$ is polynomial-time uniform. In particular, the assumption that our encoding scheme is reasonable allows a polynomial-time DTM to determine which of the two cases holds for a given string x, outputting either x itself or an encoding of Q_0. It is immediate from the definition of QCIRCUIT-ACC that $\Pr(R_x \text{ accepts}) \geq 2/3$ for every yes-input x and $\Pr(R_x \text{ accepts}) \leq 1/3$ for every no-input x, so QCIRCUIT-ACC $\in$ BQP by Definition 5.3.

Now let $A = (A_{\text{yes}}, A_{\text{no}})$ be any promise problem in BQP and take

$$\{S_x : x \in \Sigma^*\}$$

to be a polynomial-time uniform family of quantum circuits for A as in Definition 5.3. That is, each S_x has no input qubits and one output qubit, and

$$\begin{aligned} x \in A_{\text{yes}} &\implies \Pr(S_x \text{ accepts}) \geq 2/3, \\ x \in A_{\text{no}} &\implies \Pr(S_x \text{ accepts}) \leq 1/3. \end{aligned}$$

The polynomial-time uniformity of this family means precisely that the function

$$f(x) = \langle S_x \rangle$$

is computable in polynomial time. The two implications immediately above imply that $f(x)$ is a yes-input of QCIRCUIT-ACC for every $x \in A_{\text{yes}}$, while $f(x)$ is a no-input of QCIRCUIT-ACC for every $x \in A_{\text{no}}$. Thus, f itself is a Karp reduction from A to QCIRCUIT-ACC. As $A \in$ BQP was arbitrary, it follows that QCIRCUIT-ACC is BQP-hard, and therefore BQP-complete.

Solution to Problem 4

We begin with the containment PQP $\subseteq$ PP. Suppose $A = (A_{\text{yes}}, A_{\text{no}}) \in$ PQP, and let $\{Q_x : x \in \Sigma^*\}$ be a polynomial-time uniform family of quantum circuits as in the definition of PQP. By Corollary 6.16 there exist a GapP function h and a polynomial resource bound s such that

$$h(x) = 2^s \Pr(Q_x \text{ accepts})$$

for every $x \in \Sigma^*$, and therefore

$$f(x) = 2h(x) - 2^s$$

defines a GapP function (just like in the proof of Theorem 6.17). The sign of $f(x)$ agrees with the sign of

$$\Pr(Q_x \text{ accepts}) - \tfrac{1}{2},$$

so $f(x) > 0$ for every $x \in A_{\text{yes}}$ and $f(x) < 0$ for every $x \in A_{\text{no}}$, and therefore $A \in \text{PP}$ by Proposition 6.8.

In short, this is essentially the same proof as for BQP $\subseteq$ PP (Theorem 6.17), which does not actually require the assumption of bounded error.

It remains to prove the reverse containment. Suppose $A = (A_{\text{yes}}, A_{\text{no}}) \in \text{PP}$, and take a polynomial resource bound p and a language $B \in \text{P}$ as in Definition 2.18, so that these implications are true.

$$x \in A_{\text{yes}} \implies \big|\{y \in \{0,1\}^p \,:\, \langle x, y\rangle \in B\}\big| > \tfrac{1}{2} \cdot 2^p \tag{3}$$

$$x \in A_{\text{no}} \implies \big|\{y \in \{0,1\}^p \,:\, \langle x, y\rangle \in B\}\big| < \tfrac{1}{2} \cdot 2^p \tag{4}$$

Proceeding exactly as in the proof of Proposition 5.4 (BPP $\subseteq$ BQP), we may conclude that there exists a polynomial-time uniform family $\{Q_x \,:\, x \in \Sigma^*\}$ of quantum circuits, each having no input qubits and one output qubit, for which

$$\Pr(Q_x \text{ accepts}) = \frac{\big|\{y \in \{0,1\}^p \,:\, \langle x, y\rangle \in B\}\big|}{2^p}$$

for every $x \in \Sigma^*$. Specifically, each Q_x generates y uniformly at random and simulates a Boolean circuit deciding B on the pair $\langle x, y\rangle$. By the implications (3) and (4) we conclude that

$$\begin{aligned} x \in A_{\text{yes}} &\implies \Pr(Q_x \text{ accepts}) > 1/2 \\ x \in A_{\text{no}} &\implies \Pr(Q_x \text{ accepts}) < 1/2, \end{aligned}$$

so $A \in \text{PQP}$.